EXPERIMENT 05

```
clc;
clear;
close all;

% Electromagnetic Simulation and S-Parameter Characterization
% of a Microwave Directional Coupler and Two-Way Power Divider

%% 1. Operating Parameters
f0 = 10e9;              % Center frequency = 10 GHz
f = linspace(8e9,12e9,501);
fGHz = f/1e9;
BW = 2e9;

x = (f-f0)/BW;

%% 2. Directional Coupler Model
% Illustrative frequency-dependent S-parameter model

kappa = 0.3162 .* exp(-(x/0.65).^2);
through = sqrt(1-kappa.^2);
reflection = 0.025 + 0.08*abs(x);
isolation = 0.025 + 0.25*abs(x).^2;

reflection = min(reflection,0.95);
isolation = min(isolation,0.95);

% S-parameters in dB
S11_dB = 20*log10(max(reflection,1e-6));
S21_dB = 20*log10(max(through,1e-6));
S31_dB = 20*log10(max(kappa,1e-6));
```

```
S41_dB = 20*log10(max(isolation,1e-6));

% Coupler performance
coupling_dB = -S31_dB;
insertionLoss_dB = -S21_dB;
isolation_dB = -S41_dB;
directivity_dB = isolation_dB - coupling_dB;
returnLoss_dB = -S11_dB;

%% 3. Two-Way Power Divider Model

S11_div = 0.035 + 0.07*abs(x);
S21_div = (1/sqrt(2))*exp(-0.04*x.^2);
S31_div = (1/sqrt(2))*exp(-0.04*x.^2);
S23_div = 0.025 + 0.12*abs(x);

% Assumed response after dimension tuning
S11_before = S11_div;
S21_before = S21_div;

S11_after = 0.012 + 0.025*abs(x);
S21_after = (1/sqrt(2))*exp(-0.015*x.^2);

% Convert to dB
S11div_dB = 20*log10(max(S11_div,1e-6));
S23div_dB = 20*log10(max(S23_div,1e-6));

S11before_dB = 20*log10(max(S11_before,1e-6));
S11after_dB = 20*log10(max(S11_after,1e-6));

S21before_dB = 20*log10(max(S21_before,1e-6));
```

```
S21after_dB = 20*log10(max(S21_after,1e-6));

% Output power percentages
Pout1 = abs(S21_div).^2*100;
Pout2 = abs(S31_div).^2*100;

%% 4. Create ONE Large Figure with SIX Graphs

figure('Color','w', ...
    'Name','Microwave Coupler and Divider Analysis', ...
    'Units','normalized', ...
    'OuterPosition',[0 0 1 1]);

tiledlayout(2,3, ...
  'TileSpacing','loose', ...
  'Padding','loose');

%% GRAPH 1: Directional Coupler S-Parameters
nexttile;

plot(fGHz,S11_dB,'k','LineWidth',2);
hold on;
plot(fGHz,S21_dB,'b','LineWidth',2);
plot(fGHz,S31_dB,'r','LineWidth',2);
plot(fGHz,S41_dB,'g','LineWidth',2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('1. Directional Coupler S-Parameters');
legend('S_{11}','S_{21}','S_{31}','S_{41}', ...
```

```
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% GRAPH 2: Coupling, Isolation and Directivity
nexttile;

plot(fGHz,coupling_dB,'b','LineWidth',2);
hold on;
plot(fGHz,isolation_dB,'r','LineWidth',2);
plot(fGHz,directivity_dB,'m','LineWidth',2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Value (dB)');
title('2. Coupling, Isolation and Directivity');
legend('Coupling','Isolation','Directivity', ...
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% GRAPH 3: Insertion Loss and Return Loss
nexttile;

plot(fGHz,insertionLoss_dB,'b','LineWidth',2);
hold on;
plot(fGHz,returnLoss_dB,'r','LineWidth',2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');
title('3. Insertion and Return Loss');
legend('Insertion Loss','Return Loss', ...
```

```
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% GRAPH 4: Power Division Ratio
nexttile;

plot(fGHz,Pout1,'b','LineWidth',2);
hold on;
plot(fGHz,Pout2,'r--','LineWidth',2);
yline(50,'k:','Ideal 50% split','LineWidth',1.5);

grid on;
xlabel('Frequency (GHz)');
ylabel('Output Power (% of input)');
title('4. Two-Way Power Division');
legend('Output Port 2','Output Port 3', ...
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% GRAPH 5: Divider Matching and Output Isolation
nexttile;

plot(fGHz,S11div_dB,'k','LineWidth',2);
hold on;
plot(fGHz,S23div_dB,'m','LineWidth',2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('5. Divider Matching and Isolation');
legend('Input Reflection S_{11}', ...
```

```
    'Output Coupling S_{23}', ...
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% GRAPH 6: Performance Before and After Tuning
nexttile;

plot(fGHz,S11before_dB,'r--','LineWidth',2);
hold on;
plot(fGHz,S11after_dB,'r','LineWidth',2);
plot(fGHz,S21before_dB,'b--','LineWidth',2);
plot(fGHz,S21after_dB,'b','LineWidth',2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('6. Performance Before and After Tuning');
legend('S_{11} Before','S_{11} After', ...
    'S_{21} Before','S_{21} After', ...
    'Location','best');
set(gca,'FontSize',12,'LineWidth',1.2);

%% 5. Overall Figure Title
sgtitle('Microwave Directional Coupler and Two-Way Power Divider', ...
     'FontSize',18,'FontWeight','bold');

%% 6. Display Numerical Results at 10 GHz
[~,idx] = min(abs(f-f0));

fprintf('\n--- Directional Coupler at %.2f GHz ---\n',f0/1e9);
fprintf('Coupling Factor  : %.2f dB\n',coupling_dB(idx));
```

```
fprintf('Isolation       : %.2f dB\n',isolation_dB(idx));
fprintf('Directivity     : %.2f dB\n',directivity_dB(idx));
fprintf('Insertion Loss  : %.2f dB\n',insertionLoss_dB(idx));
fprintf('Return Loss     : %.2f dB\n',returnLoss_dB(idx));

fprintf('\n--- Two-Way Power Divider at %.2f GHz ---\n',f0/1e9);
fprintf('Output Port 2 Power : %.2f %%\n',Pout1(idx));
fprintf('Output Port 3 Power : %.2f %%\n',Pout2(idx));
fprintf('Ideal Equal Split   : 50 %% per output\n');

disp('Simulation completed: all six graphs are in one figure.');
```

